

Group 2 Final Project: DataCo Supply Chain

Data Dictionary

Original Dataset Source: [DataCo Smart Supply Chain](#)

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Overview

This document contains both the data dictionary and business rules of the project. The data dictionary documents all entities/tables, columns/attributes, data types, constraints, and definitions for the DataCo Smart Supply Chain relational database. It is intended to serve as the authoritative reference for ER modeling and relational database creation.

The database consists of 11 entities derived from the DataCo Smart Supply Chain dataset. Column names have been standardized to uppercase with underscores. Several columns were renamed, removed, or restructured from its original form during data preparation to improve clarity and eliminate redundancy.

Table Summary

#	Table	Type	Description
1	CUSTOMER	Core	Customers who place orders
2	DEPARTMENT	Lookup	Store departments managing product categories
3	CATEGORY	Lookup	Product categories grouping related products
4	PRODUCT	Core	Products available in the supply chain catalog
5	SHIPPING_MODE	Lookup	Available shipping methods for fulfilling orders
6	PAYMENT_TYPE	Lookup	Accepted payment transaction types
7	MARKET	Lookup	Global markets to which orders are delivered
8	ORDER_LOCATION	Core	Delivery destination details for orders
9	ORDER	Core	Customer orders placed through the system
10	LINE	Junction	Individual line items linking orders to products
11	SHIPMENT	Core	Shipment and delivery details per order

Table Definitions

1. CUSTOMER

Stores information about customers who place orders through the supply chain system.

Column Name	Data Type	Constraints	Description
CUSTOMER_ID	INT	PK, NOT NULL	Unique identifier for each customer
CUSTOMER_FNAME	VARCHAR(50)	NOT NULL	Customer's first name
CUSTOMER_LNAME	VARCHAR(50)	NOT NULL	Customer's last name
CUSTOMER_SEGMENT	VARCHAR(50)	NOT NULL	Market segment the customer belongs to: Consumer, Corporate, or Home Office
CUSTOMER_STREET	VARCHAR(100)	NOT NULL	Street address of the store where the purchase is registered
CUSTOMER_CITY	VARCHAR(50)	NOT NULL	City where the customer made the purchase
CUSTOMER_STATE	VARCHAR(50)	NOT NULL	State where the store where the purchase is registered belongs
CUSTOMER_ZIPCODE	VARCHAR(10)	NULL	Customer zip code. Nullable — not always available
CUSTOMER_COUNTRY	VARCHAR(50)	NOT NULL	Country where the customer made the purchase

2. DEPARTMENT

Represents the store departments that oversee and manage product categories.

Column Name	Data Type	Constraints	Description
DEPARTMENT_ID	INT	PK, NOT NULL	Unique department code for the store department
DEPARTMENT_NAME	VARCHAR(50)	NOT NULL	Name of the store department

3. CATEGORY

Represents product categories that group related products together, each belonging to a department.

Column Name	Data Type	Constraints	Description
PRODUCT_CATEGORY_ID	INT	PK, NOT NULL	Unique product category code
CATEGORY_NAME	VARCHAR(50)	NOT NULL	Description/name of the product category
DEPARTMENT_DEPARTMENT_ID	INT	FK → DEPARTMENT, NOT NULL	Foreign key referencing the department this category belongs to

4. PRODUCT

Stores product information available in the supply chain catalog.

Column Name	Data Type	Constraints	Description
PRODUCT_ID	INT	PK, NOT NULL	Unique product code (sourced from Product Card Id — RFID-based product identifier)
PRODUCT_NAME	VARCHAR(100)	NOT NULL	Name of the product
PRODUCT_PRICE	DECIMAL(10,2)	NOT NULL	Listed price of the product
PRODUCT_STATUS	TINYINT(1)	NOT NULL, DEFAULT 0	Stock availability flag: 0 = product is available, 1 = product is not available
CATEGORY_PRODUCT_CATEGORY_ID	INT	FK → CATEGORY, NOT NULL	Foreign key referencing the product category this product belongs to

5. SHIPPING_MODE

Look-up table for the available shipping methods used to fulfill orders.

Column Name	Data Type	Constraints	Description
SHIPPING_MODE_ID	INT	PK, NOT NULL, AUTO_INCREMENT	Surrogate key — auto-generated unique identifier for each shipping mode
SHIPPING_MODE	VARCHAR(20)	NOT NULL	Shipping method used for the order: Standard Class, First Class, Second Class, or Same Day

6. PAYMENT_TYPE

Lookup table for the types of payment transactions accepted in the system.

Column Name	Data Type	Constraints	Description
PAYMENT_TYPE_ID	INT	PK, NOT NULL, AUTO_INCREMENT	Surrogate key — auto-generated unique identifier for each payment type
PAYMENT_TYPE	VARCHAR(20)	NOT NULL	Type of payment transaction made: Debit, Transfer, Cash, or Payment

7. MARKET

Lookup table representing the global markets to which orders are delivered.

Column Name	Data Type	Constraints	Description
MARKET_ID	INT	PK, NOT NULL, AUTO_INCREMENT	Surrogate key — auto-generated unique identifier for each market
MARKET	VARCHAR(20)	NOT NULL	Market region to which the order is delivered: Africa, Europe, LATAM, Pacific Asia, or USCA

8. ORDER_LOCATION

Stores the delivery destination details for orders, grouped by geographic location and market.

Column Name	Data Type	Constraints	Description
LOCATION_ID	INT	PK, NOT NULL, AUTO_INCREMENT	Surrogate key — auto-generated unique identifier for each order delivery location
ORDER_CITY	VARCHAR(50)	NOT NULL	Destination city of the order
ORDER_STATE	VARCHAR(50)	NOT NULL	State of the region where the order is delivered
ORDER_COUNTRY	VARCHAR(50)	NOT NULL	Destination country of the order
ORDER_REGION	VARCHAR(50)	NOT NULL	Region of the world where the order is delivered (e.g. Southeast Asia, Western Europe, West of USA)
ORDER_ZIPCODE	VARCHAR(10)	NULL	Zip code of the order destination. Nullable — approximately 96% of records do not have this value
MARKET_MARKET_ID	INT	FK → MARKET, NOT NULL	Foreign key referencing the market this location belongs to

9. ORDER

Represents customer orders placed through the supply chain system, linking customers, payment, shipping mode, and delivery location.

Column Name	Data Type	Constraints	Description
ORDER_ID	INT	PK, NOT NULL	Unique order code
ORDER_DATE	DATETIME	NOT NULL	Date and time on which the order was placed
ORDER_STATUS	VARCHAR(20)	NOT NULL	Current status of the order: COMPLETE, PENDING, CLOSED, PENDING_PAYMENT, CANCELED, PROCESSING, SUSPECTED_FRAUD, ON_HOLD, or PAYMENT_REVIEW

ORDER_PROFIT	DECIMAL(10,2)	NOT NULL	Profit generated per order
CUSTOMER_CUSTOMER_ID	INT	FK → CUSTOMER, NOT NULL	Foreign key referencing the customer who placed the order
SHIPPING_MODE_SHIPPING_MODE_ID	INT	FK → SHIPPING_MODE, NOT NULL	Foreign key referencing the shipping mode assigned to the order
ORDER_LOCATION_LOCATION_ID	INT	FK → ORDER_LOCATION, NOT NULL	Foreign key referencing the delivery location of the order
PAYMENT_TYPE_PAYMENT_TYPE_ID	INT	FK → PAYMENT_TYPE, NOT NULL	Foreign key referencing the payment type used for the order

10. LINE

Junction/bridge table representing individual line items within an order. Each record links one order to one product and captures the transaction details for that specific item.

Column Name	Data Type	Constraints	Description
LINE_ID	INT	PK, NOT NULL	Unique order line item code
ORDER_ORDER_ID	INT	FK → ORDER, NOT NULL	Foreign key referencing the order this line belongs to
PRODUCT_PRODUCT_ID	INT	FK → PRODUCT, NOT NULL	Foreign key referencing the product included in this line
LINE_QUANTITY	INT	NOT NULL	Number of product units included in this order line
LINE_DISCOUNT	DECIMAL(10,2)	NOT NULL	Discount amount applied to this order line item
LINE_DISCOUNT_RATE	DECIMAL(5,4)	NOT NULL	Discount rate (percentage) applied to this order line item
LINE_PRODUCT_PRICE	DECIMAL(10,2)	NOT NULL	Price of the product before any discount is applied
LINE_PROFIT_RATIO	DECIMAL(5,4)	NOT NULL	Profit ratio for this order line item
LINE_SALES	DECIMAL(10,2)	NOT NULL	Gross sales value for this line item before discount (LINE_QUANTITY × LINE_PRODUCT_PRICE)

LINE_TOTAL	DECIMAL(10,2)	NOT NULL	Net total amount for this line item after discount is applied (LINE_SALES × (1 – LINE_DISCOUNT_RATE))
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11. SHIPMENT

Stores shipment and delivery details for each order. Has a one-to-one relationship with ORDER.

Column Name	Data Type	Constraints	Description
SHIPMENT_ID	INT	PK, NOT NULL, AUTO_INCREMENT	Surrogate key — auto-generated unique identifier for each shipment record
ORDER_ORDER_ID	INT	FK → ORDER, NOT NULL, UNIQUE	Foreign key referencing the associated order. UNIQUE constraint enforces the one-to-one relationship with ORDER
SHIPPING_DATE	DATETIME	NOT NULL	Exact date and time of shipment
DAYS_FOR_SHIPPING_REAL	INT	NOT NULL	Actual number of shipping days taken for the purchased product to be delivered
DAYS_FOR_SHIPMENT_SCHEDULED	INT	NOT NULL	Originally scheduled number of days for delivery of the purchased product
DELIVERY_STATUS	VARCHAR(25)	NOT NULL	Delivery status of the order: Advance Shipping, Late Delivery, Shipping Canceled, or Shipping on Time

LATE_DELIVERY_RISK	TINYINT(1)	NOT NULL, DEFAULT 0	Binary flag indicating whether the shipment is at risk of late delivery: 0 = not late, 1 = late
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Relationship Summary

From Entity	Verb	To Entity	Cardinality
CUSTOMER	places	ORDER	One-to-Many
ORDER	contains	LINE	One-to-Many
PRODUCT	is included in	LINE	One-to-Many
CATEGORY	classifies	PRODUCT	One-to-Many
DEPARTMENT	manages	CATEGORY	One-to-Many
ORDER	generates	SHIPMENT	One-to-One
SHIPPING_MODE	defines	ORDER	One-to-Many
PAYMENT_TYPE	processes	ORDER	One-to-Many
MARKET	encompasses	ORDER_LOCATION	One-to-Many
ORDER_LOCATION	receives	ORDER	One-to-Many